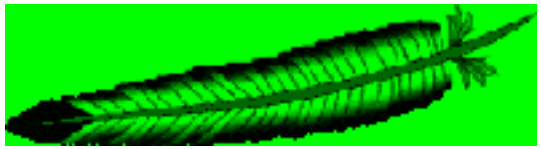
WE03

Troubleshooting Apache configurations



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blurb!

There are but very few personal problems that can’t be solved by application of an appropriate amount of dynamite

Does this apply to Apache?

Agenda

- 1. Introduction
- 2. Tools
- 3. Logs
- 4. Troubleshooting

Introduction - #Apache

Doesn’t work!

Look, buddy, "doesn’t work" is a strong statement.

Does it sit on the couch all day?

Is it making faces at you?

Does it want more money?

Please be specific!

Introduction - asking questions

What, exactly, is happening, and how does this differ from what you expect to be happening?

Browsers lie

- Internet explorer
  - own error messages
  - dns error
  - ...
- Mozilla & friends
  - agressive caching
  - ...
- The server
  - generic errors
  - security

Tools

- 0. httpd
- 1. telnet
- 2. openssl s_client
- 3. netstat
- 4. Other tools

Tools - httpd

Not really a tool, but

httpd

- -V : show compile settings
- -t : run syntax check for config files
- -l : list compiled in modules
- -L : list available configuration directives
- -S : a synonym for -t -D DUMP_VHOSTS

Tools - httpd -V

Server version: Apache/2.1.0-dev

Server built: Oct 4 2003 13:02:55
Server's Module Magic Number: 20030821:0

Architecture: 32-bit
Server compiled with....

-D APACHE_MPM_DIR="server/mpm/worker"

-D APR_HAS_SENDFILE

-D APR_HAS_MMAP

-D APR_HAVE_IPV6 (IPv4-mapped addresses enabled)

-D HTTPD_ROOT="/opt/httpd-2.1"

-D SUEXEC_BIN="/opt/httpd-2.1/bin/suexec"

-D DEFAULT_SCOREBOARD="logs/apache_runtime_status"

-D DEFAULT_ERRORLOG="logs/error_log"

-D AP_TYPES_CONFIG_FILE="conf/mime.types"

-D SERVER_CONFIG_FILE="conf/httpd.conf"

Tools - telnet

$ telnet www.apache.org 80

Trying 209.237.227.195...
Connected to www.apache.org.
Escape character is '^]'.

HEAD / HTTP/1.0

HTTP/1.1 200 OK
Date: Thu, 02 Oct 2003 19:21:13 GMT
Server: Apache/2.0.48-dev (Unix)
Accept-Ranges: bytes
Cache-Control: max-age=86400
Expires: Fri, 03 Oct 2003 19:21:13 GMT
Content-Length: 10036
Connection: close
Content-Type: text/html; charset=ISO-8859-1

Connection closed by foreign host.

Tools - telnet

$ telnet www.apache.org 80

Trying 209.237.227.195...
Connected to www.apache.org.
Escape character is '^]'.

GET /foundation HTTP/1.0
Host: www.apache.org

HTTP/1.1 301 Moved Permanently
Date: Fri, 03 Oct 2003 09:49:35 GMT
Server: Apache/2.0.48-dev (Unix)
Location: http://www.apache.org/foundation/
Content-Length: 325
Connection: close
Content-Type: text/html; charset=iso-8859-1

<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">

Tools - openssl s_client

openssl s_client -connect 127.0.0.1:443

-connect host:port

-showcerts
-state
-debug

-cert certname -key keyfile -CAfile file

-ssl2, -ssl3, -tls1, -no_ssl2, -no_ssl3, -no_tls1

-cipher cipherlist

-reconnect

Tools - openssl s_client

ex1:
$ openssl s_client -connect www.apache.org:443

```
connect: Connection refused
connect:errno=29
```

ex2:
openssl s_client -connect HOST:PORT -debug

```
CONNECTED(00000003)
write to 08158160 [081581A8] (130 bytes => 130 (0x82))
0000 - 80 80 01 03 01 00 57 00-00 00 20 00 00 16 00 00   ......W... .....
0010 - 13 00 00 0a 07 00 c0 00-00 66 00 00 07 00 00 05   .........f......
0020 - 00 00 04 05 00 80 03 00-80 01 00 80 08 00 80 00   ................
0030 - 00 65 00 00 64 00 00 63-00 00 62 00 00 61 00 00   .e..d..c..b..a..
0040 - 60 00 00 15 00 00 12 00-00 09 06 00 40 00 00 14   `...........@...
0050 - 00 00 11 00 00 08 00 00-06 00 00 03 04 00 80 02   ................
0060 - 00 80 1c 42 8e db df 85-34 b5 f6 4c 1e e7 cb 57   ...B....4..L...W
0070 - b2 ef 99 21 91 92 1e 08-3f 75 84 16 ce f7 88 63   ...!....?u.....c
0080 - f4 5a                                             .Z
read from 08158160 [0815D708] (7 bytes => 7 (0x7))
0000 - 3c 21 44 4f 43 54 59                              <!DOCTY
17300:error:140770FC:SSL routines:SSL23_GET_SERVER_HELLO:unknown protocol:s23_clnt.c:470:
```

Tools - netstat

```
$ netstat -an
Active Internet connections (servers and established)
Proto Recv-Q Send-Q     Local Address         Foreign Address       State
tcp        0      0     0.0.0.0:80            0.0.0.0:*             LISTEN
tcp        0      0     0.0.0.0:21            0.0.0.0:*             LISTEN
tcp        0      0     0.0.0.0:22            0.0.0.0:*             LISTEN
```

```
$ netstat -an| grep LISTEN
```

```
$ netstat -anp --inet
```

Tools - Other

See http://httpd.apache.org/dev/debugging.html

- gdb
- strace/trace/truss
- tcpdump

See http://www.research.att.com/~hpk/wsp/

- WSP

Logs - Always check the error log

- ErrorLog - (logs/error_log)
  - LogLevel - (debug, info, notice, warn, error...)
- RewriteLog
  - RewriteLogLevel (0 - 9)
- ScriptLog
  - ScriptLogBuffer (1024)
  - ScriptLogLength
- --with-suexec-log= - (logs/suexec_log)
  - AP_LOG_EXEC (suexec -V)
- SSLLog - 1.3 mod_ssl only
  - SSLLogLevel - 1.3 mod_ssl only

Troubleshooting

- 1. Connection errors
- 2. Startup errors
- 3. 404s
- 4. 403s
- 5. Redirection errors
- 6. Scripting errors

1. Connection errors

1. is the server running at all
   - ps
   - netstat -an
   - telnet 127.0.0.1 80

2. did the request reach the server
   - tail -f logs/access
   - tcpdump

3. protocol errors (SSL)
   - openssl s_client

2. Startup errors

- Always look in the error log
- strace/truss/trace

"Socket Already in use"

Common causes

- ServerName
- logfiles in directories that doesn’t exist
- Segfault

3. 404s

404 not found

- error log
- access log

Common causes

- (Script)Alias - (also mismatched /)
- Redirects
  - telnet
  - ServerName
- Relative / Absolute paths
- RewriteRules
  - RewriteLog / RewriteLogLevel 9

4. 403s

403 Forbidden

- error log

"client denied by server configuration"

- httpd.conf
  - deny from

"user dfsfdd not found"

- check password file

"does not meet ’require’ments for user/valid-user"

- Check Require statement
  - valid-user / group / user

5. Redirection errors

See also 404s

6. Scripting errors

500 Internal server error

- error log
- execute script locally - ./script
- suexec log
- ScriptLog

No output / unparsed output

- ScriptAlias
- Options +ExecCGI

7. htaccess errors

500 Internal server error
- error log
- "syntax error" -> 500
- Not allowed in this context
- AllowOverride

"No effect"
- AllowOverride
- echo "garbage" >> .htaccess

Questions?

mads@apache.org

#apache @ irc.freenode.net

This presenation has ended.
Click anywhere to return.

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